

AMAZON SALES ANALYTICS DASHBOARD

Prepared By

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Project Overview

The Amazon Sales Analytics Dashboard project was developed to analyze sales performance, customer behavior, product categories, discounts, ratings, and regional trends using SQL and Power BI.

The primary objective of this project is to convert raw sales data into meaningful business insights that can support better decision-making and improve overall business performance.

Business Problem

Organizations generate large volumes of sales data every day. However, without proper analysis, it is difficult to identify top-performing products, customer preferences, sales trends, and revenue-driving factors.

This project aims to solve these challenges by analyzing Amazon sales data and presenting insights through interactive dashboards.

Project Objectives

- Analyze total revenue and sales performance.
 - Identify top-performing product categories.
 - Analyze monthly revenue trends.
 - Evaluate customer ratings and reviews.
 - Measure the impact of discounts on revenue.
 - Compare regional sales performance.
 - Create an interactive dashboard for business reporting.
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Tools and Technologies Used

- PostgreSQL
 - SQL
 - Power BI
 - Power Query
 - DAX
 - Microsoft Excel
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Dataset Description

The dataset contains Amazon sales transaction information including:

- Product ID
- Product Category
- Order Date
- Customer Region
- Revenue
- Discount Percentage
- Rating
- Review Count
- Payment Method

SQL Analysis

Revenue by Product Category

Calculated total revenue generated by each product category to identify the highest-performing categories.

	product_category text	total_revenue double precision
1	Electronics	4151519.5700000115
2	Beauty	4148708.0499999993
3	Fashion	4145638.110000002
4	Home & Kitchen	4103456.1000000183
5	Books	4092901.3400000036
6	Sports	4049150.7399999994

Monthly Revenue Trend Analysis

Analyzed month-wise revenue performance to understand growth patterns.

	month timestamp without time zone	monthly_revenue double precision
1	2023-02-01 00:00:00	949447.7699999991
2	2022-11-01 00:00:00	949924.2199999993
3	2022-02-01 00:00:00	973337.6200000013
4	2023-12-01 00:00:00	986253.9200000003
5	2023-04-01 00:00:00	989342.0199999996

Regional Revenue Ranking

Ranked customer regions based on total revenue contribution using SQL Window Functions.

	customer_region text	total_revenue double precision	rank bigint
1	Middle East	8301844.500000002	1
2	North America	8277217.8399999947	2
3	Europe	8112311.569999994	3

Payment Method Analysis

Examined customer payment preferences across different regions.

	customer_region text	payment_method text	usage_method bigint
1	Europe	Debit Card	2551
2	Europe	Wallet	2544
3	Europe	UPI	2478
4	Europe	Credit Card	2468
5	Europe	Cash on Delivery	2411

Customer Rating Analysis

Highest ratings across customer regions.

	customer_region text	avg_rating numeric
1	Middle East	3.02
2	North America	3.00
3	Europe	2.97

Discount Impact Analysis

Analyzed the impact of discount levels on overall revenue.

	discount_category text	total_orders bigint	avg_revenue numeric
1	Low Discount (0-14%)	18839	723.25
2	Medium Discount (15-29...	12420	629.78
3	High Discount (30%+)	6215	522.00

Product Performance Analysis

Identified products with high review counts but low ratings.

	product_id bigint	product_category text	rating double precision	review_count bigint	total_revenue double precision
1	3399	Fashion	1.6	401	169.48
2	4405	Books	2.6	401	661.76
3	1686	Electronics	1.5	401	331.9
4	3815	Electronics	1.2	401	334.32
5	4040	Sports	2.1	401	731.42

Revenue Classification Analysis

Classified product categories into High Revenue and Low Revenue segments.

	product_category text	revenue double precision	revenue_type text
1	Electronics	4151519.5700000115	HIGH REVEN...
2	Beauty	4148708.0499999993	HIGH REVEN...
3	Fashion	4145638.110000002	HIGH REVEN...
4	Home & Kitchen	4103456.1000000183	HIGH REVEN...
5	Books	4092901.3400000036	LOW REVENUE
6	Sports	4049150.7399999994	LOW REVENUE

Top Categories by Region

Identified the top three product categories within each customer region.

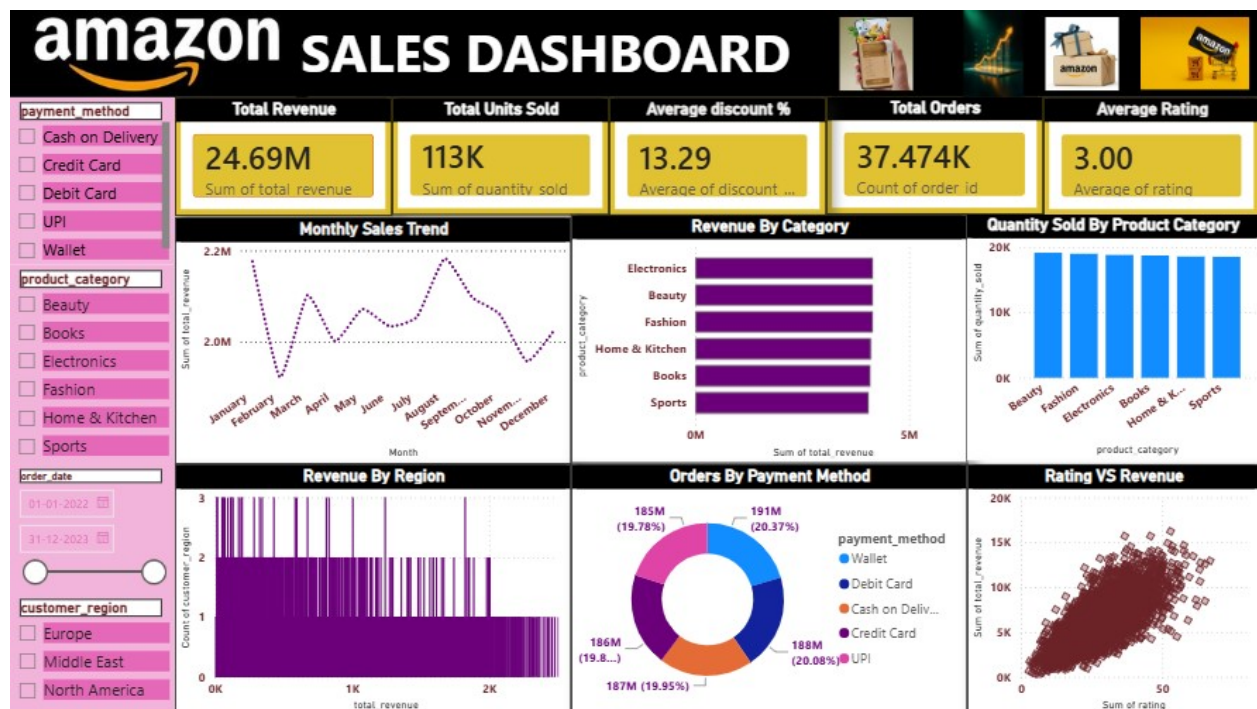
	customer_region text	product_category text	total_revenue double precision	rank bigint
1	Europe	Electronics	1407118.9399999974	1
2	Europe	Fashion	1366107.0000000002	2
3	Europe	Beauty	1358226.4199999978	3
4	Middle East	Books	1428560.0700000022	1
5	Middle East	Fashion	1390228.600000001	2

Power BI Dashboard

Dashboard Page 1 – Executive Overview

Features:

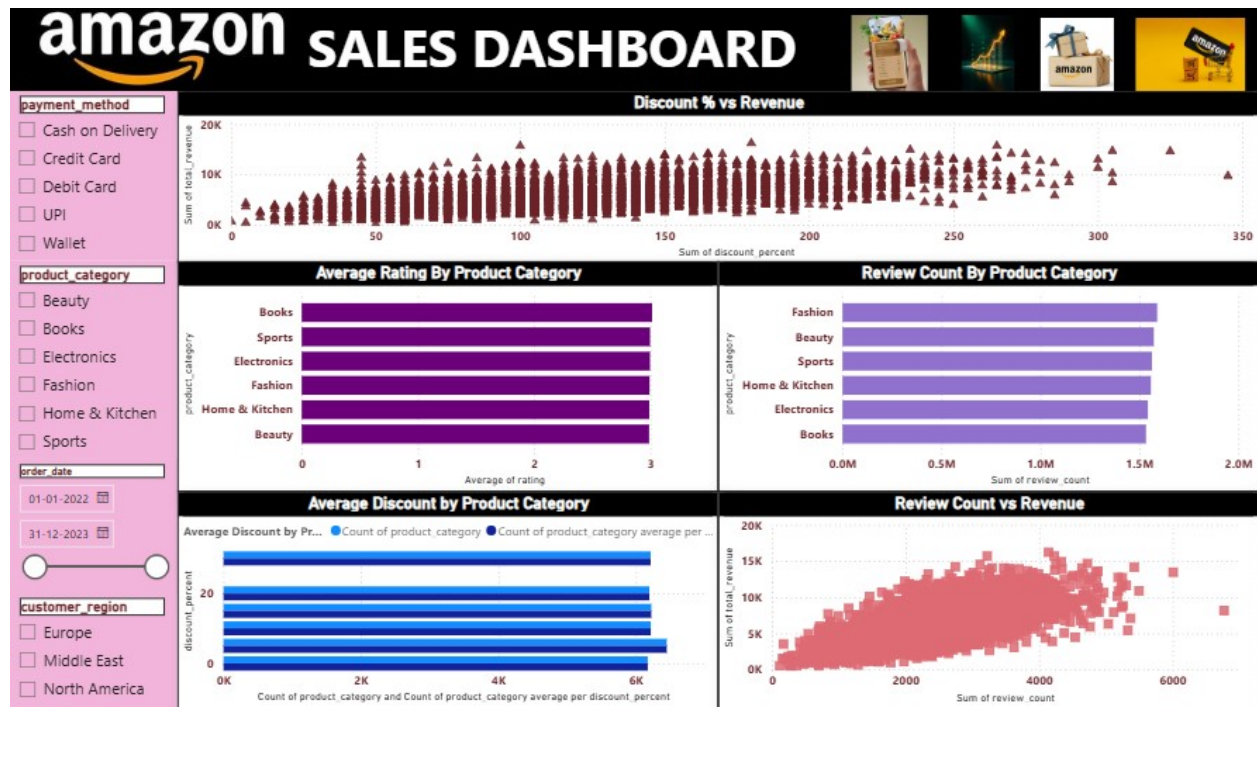
- Total Revenue
- Total Orders
- Total Units Sold
- Average Rating
- Revenue by Product Category
- Revenue by Region
- Monthly Revenue Trend



Dashboard Page 2 – Customer & Discount Analysis

Features:

- Discount % vs Revenue
- Average Rating by Product Category
- Review Count by Product Category
- Review Count vs Revenue
- Average Discount by Product Category



Key Insights

- Product categories contribute differently to overall revenue.
- Customer purchasing behavior varies across regions.
- Discounts have a measurable impact on revenue generation.
- Customer ratings influence product performance.
- Revenue trends help identify seasonal sales patterns.

Conclusion

The Amazon Sales Analytics Dashboard successfully transformed raw sales data into actionable business insights using SQL and Power BI. This project demonstrates practical skills in data analysis, data visualization, dashboard development, SQL querying, and business intelligence reporting.